

SmartCare AI: An Explainable, Comorbidity-Aware Multi-Task Machine Learning Framework for Early Chronic Disease Risk Stratification and Clinical Decision Support

SmartCare AI Research & Engineering Consortium

Department of Computer Science & Engineering • Healthcare Informatics Lab • Clinical Intelligence Group
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Abstract—Chronic non-communicable diseases—specifically Type 2 Diabetes (T2D), Cardiovascular Disease (CVD), and Chronic Kidney Disease (CKD)—represent the leading causes of global mortality and healthcare expenditure. In clinical reality, these conditions share mutual pathophysiology and compounding vascular damage. Conventional single-disease diagnostic models fail to capture these intricate comorbidity synergies and operate as siloed 'black boxes.' In this paper, we propose *SmartCare AI*, a unified clinical machine learning framework that delivers comorbidity-aware multi-task risk stratification, rigorous probability calibration, game-theoretic explainability, and evidence-based wellness interventions. Our architecture harmonizes heterogeneous clinical datasets (BRFSS 2015 N=70,692; Kaggle CVD N=68,205; UCI CKD N=400) into a standardized 30-dimensional shared feature representation. We enforce strict leakage-free preprocessing with training-fitted median imputation and RobustScaler transformations. A shared neural encoder network with task-masked loss and task-balanced batch sampling (20% CKD allocation) is jointly trained across all three conditions and calibrated via Platt sigmoid scaling. On strictly isolated, held-out test splits, SmartCare AI achieves robust performance: T2D (ROC-AUC: 0.8276, Recall: 88.72%, F1: 0.7751, Brier: 0.1682), CVD (ROC-AUC: 0.7974, Recall: 79.13%, F1: 0.7306, Brier: 0.1827), and CKD (ROC-AUC: 0.9953, Recall: 94.59%, F1: 0.9589, Brier: 0.0490). We benchmark against Logistic Regression, Random Forest, and XGBoost baselines on identical test partitions. Game-theoretic SHAP explainability exposes global biomarker rankings and patient-level attributions, parameterizing automated clinical guidelines and longitudinal risk alerts within an enterprise FastAPI microservice layer and MongoDB database. SmartCare AI establishes a reproducible, scientifically grounded benchmark for next-generation clinical decision support systems.

Index Terms—Comorbidity-Aware Prediction, Multi-Task Learning, Shared Feature Representation, Explainable AI (XAI), SHAP Values, Platt Calibration, Cardio-Renal-Metabolic Syndrome, Type 2 Diabetes, Cardiovascular Disease, Chronic Kidney Disease, CDSS, FastAPI, MongoDB.

I. INTRODUCTION & CLINICAL MOTIVATION

Chronic non-communicable diseases (NCDs) constitute the foremost clinical and economic burden worldwide, accounting for over 74% of all annual deaths globally according to the World Health Organization (WHO). Among these, Type 2 Diabetes Mellitus (T2D), Cardiovascular Disease (CVD), and Chronic Kidney Disease (CKD) form an interconnected, multi-system pathological triad known clinically as the *Cardio-Renal-Metabolic (CRM) Syndrome*. Endothelial dysfunction, systemic microvascular inflammation, chronic hypertension, and dyslipidemia act as mutually compounding drivers across all three organs.

Despite their intrinsic interdependencies, traditional diagnostic workflows and existing machine learning (ML) diagnostic applications predominantly address these diseases as isolated, single-label classification problems. This siloed methodology presents several critical failure modes: (1) **Comorbidity Blindness**: models trained exclusively on single disease indicators miss cross-organ risk amplifications; (2) **Uncalibrated Overconfidence**: standard ML classifiers produce raw output scores that deviate from true empirical probabilities, leading to misinformed triaging; (3) **The Explainability Gap**: complex black-box architectures prevent clinicians from verifying why a patient was classified as high risk; and (4) **Lack of Actionable Linkage**: prediction systems rarely bridge estimated risks to guideline-adherent dietary, lifestyle, or specialist intervention pathways.

To address these challenges, we introduce **SmartCare AI**: a modular, transparent, and comorbidity-aware multi-task predictive platform. SmartCare AI unifies multi-source clinical data ingestion, a 30-dimensional harmonized feature representation, joint multi-task representation learning, Platt probability calibration, game-theoretic SHAP explainability, evidence-based wellness guidance, and longitudinal monitoring into an integrated clinical decision support system (CDSS).

II. SYSTEM ARCHITECTURE & MODULAR PIPELINE

SmartCare AI is engineered as a decoupled, six-stage microservice architecture backed by an enterprise NoSQL database and asynchronous RESTful APIs:

- **Module 1 (Data Ingestion & Leakage-Free Preprocessing)**: Harmonizes multi-cohort clinical features into a unified 30-dimensional schema, performing stratified cohort splits and fitting median imputers and RobustScalers strictly on training data.
- **Module 2 (Multi-Task Learning & Risk Stratification)**: Trains a shared neural encoder network with task-specific heads, task-masked BCE loss, and task-balanced batch sampling (20% CKD allocation), followed by post-hoc Platt sigmoid probability calibration.
- **Module 3 (Explainable AI & Risk Attribution)**: Computes TreeSHAP and KernelSHAP attributions to expose global population rankings and local patient waterfall decompositions.

- **Module 4 (Personalized Wellness & Clinical Guidance):** Maps SHAP risk drivers directly into evidence-based dietary, exercise, smoking cessation, and specialist consultation protocols.
- **Module 5 (Risk Monitoring & Longitudinal CDSS):** Tracks longitudinal multi-disease trajectory alerts over time, computing biomarker velocity and triggering multi-tier clinician notifications.
- **Module 6 (Interactive Dashboard & LLM Wellness Chatbot):** Provides a responsive glassmorphism web console with multi-language support (English, Spanish, Hindi, Tamil) and a context-bounded conversational wellness assistant.

End-to-End Explainable Prediction & Wellness Guidance Flow (SmartCare AI)



Fig. 1. End-to-End Explainable Multi-Task Prediction, Feature Attribution, and Clinical Guidance Flow across SmartCare AI.

III. DATASETS & 30-FEATURE HARMONIZATION

SmartCare AI utilizes three distinct clinical datasets representing diverse demographic, behavioral, and physiological cohorts: (1) **CDC BRFSS 2015 Diabetes Cohort** (\$N = 70,692\$), a 50-50 balanced survey dataset with 21 behavioral and demographic indicators; (2) **Cardiovascular Disease Cohort** (\$N = 68,205\$), containing measured vitals and hemodynamic attributes; and (3) **UCI Chronic Kidney Disease Cohort** (\$N = 400\$), providing detailed laboratory biochemistry and urinalysis parameters.

Because these source datasets originate from separate studies and survey instruments, they observe differing subsets of variables. Rather than fabricating measurements or using invalid surrogates (e.g., treating CAD as smoking status, inferring previous heart attacks from high blood pressure, or deriving glucose from cholesterol), SmartCare AI defines an explicit **30-Dimensional Harmonized Shared Feature Schema**. Unobserved features within each cohort are represented as missing (\$\text{ext}\{\text{NaN}\}\$) during feature extraction and imputed using a training-fitted median imputation mechanism.

A. Authoritative 30-Feature Mapping Specification

#	Unified Feature	T2D Source (BRFSS)	CVD Source (Cardio)	CKD Source (UCI)	Transformation / Derivation Logic
1	age_years	Age (1-13)	age (days) / age_years	age	Midpoint years (T2D); age/365.25 (CVD); Direct (CKD)
2	gender	Sex (0=F, 1=M)	gender (1=F, 2=M)	-- (Unavailable)	Binary flag (0.0=Female, 1.0=Male); Median imputed (CKD)
3	education	Education (1-6)	-- (Unavailable)	-- (Unavailable)	Ordinal education tier (1-6); Median imputed (CVD/CKD)
4	income	Income (1-8)	-- (Unavailable)	-- (Unavailable)	Ordinal income level (1-8); Median imputed (CVD/CKD)
5	bmi	BMI	bmi / height, weight	-- (Unavailable)	Body Mass Index (kg/m ²); Median imputed (CKD)
6	high_bp	HighBP (0/1)	ap_hi, ap_lo	htn (yes/no)	Observed flag (T2D/CKD); Derived ap_hi>=140 or ap_lo>=90 (CVD)
7	ap_hi	-- (Unavailable)	ap_hi	bp	Systolic BP (mmHg); Clipped [80, 220]; Median imputed (T2D)
8	ap_lo	-- (Unavailable)	ap_lo	-- (Unavailable)	Diastolic BP (mmHg); Clipped [50, 140]; Median imputed (T2D/CKD)
9	pulse_pressure	-- (Unavailable)	ap_hi, ap_lo	-- (Unavailable)	Derived: PP = ap_hi - ap_lo; Median imputed (T2D/CKD)
10	mean_arterial_pressure	-- (Unavailable)	ap_hi, ap_lo	-- (Unavailable)	Derived: MAP = ap_lo + PP/3; Median imputed (T2D/CKD)
11	cholesterol	HighChol (0/1)	cholesterol (1-3)	-- (Unavailable)	Binary (T2D); Ordinal (CVD); Median imputed (CKD)
12	glucose	-- (Unavailable)	gluc (1-3)	bgr	Midpoint (100, 140, 200 mg/dL) from CVD; Measured BGR (CKD)
13	serum_creatinine	-- (Unavailable)	-- (Unavailable)	sc	Serum creatinine (mg/dL); Clipped [0.4, 15.0]; Median imputed (T2D/CVD)
14	blood_urea	-- (Unavailable)	-- (Unavailable)	bu	Blood urea nitrogen (mg/dL); Clipped [5.0, 200.0]; Median (T2D/CVD)
15	hemoglobin	-- (Unavailable)	-- (Unavailable)	hemo	Hemoglobin (g/dL); Clipped [3.0, 20.0]; Median (T2D/CVD)
16	albumin_level	-- (Unavailable)	-- (Unavailable)	al (0-5)	Urinary dipstick albumin (0-5); Median imputed (T2D/CVD)
17	specific_gravity	-- (Unavailable)	-- (Unavailable)	sg	Urine specific gravity (1.005-1.025); Median imputed (T2D/CVD)
18	bun_creatinine_ratio	-- (Unavailable)	-- (Unavailable)	bu, sc	Derived: bu / max(sc, 0.1), clipped at 100.0; Median (T2D/CVD)
19	egfr_proxy	-- (Unavailable)	-- (Unavailable)	sc, age	Derived MDRD proxy: $175 * sc^{(-1.154)} * age^{(-0.203)}$ (CKD)
20	anemia_flag	-- (Unavailable)	-- (Unavailable)	ane, hemo	Binary flag: ane=='yes' or hemo<12.0 g/dL; Median (T2D/CVD)
21	smoker	Smoker (0/1)	smoke (0/1)	-- (Unavailable)	Tobacco smoking status binary flag; Median imputed (CKD)
22	phys_activity	PhysActivity (0/1)	active (0/1)	-- (Unavailable)	Physical exercise indicator; Median imputed (CKD)
23	alcohol_consumption	HvyAlcoholConsump (0/1)	alco (0/1)	-- (Unavailable)	Heavy alcohol consumption flag; Median imputed (CKD)
24	fruits	Fruits (0/1)	-- (Unavailable)	-- (Unavailable)	Daily fruit consumption indicator; Median imputed (CVD/CKD)
25	veggies	Veggies (0/1)	-- (Unavailable)	-- (Unavailable)	Daily vegetable consumption indicator; Median (CVD/CKD)
26	heart_disease_or_attack	HeartDiseaseorAttack (0/1)	-- (Unavailable)	cad (yes/no)	Prior myocardial infarction / CAD history; Median imputed (CVD)
27	stroke	Stroke (0/1)	-- (Unavailable)	-- (Unavailable)	Prior stroke history binary indicator; Median imputed (CVD/CKD)
28	diff_walk	DiffWalk (0/1)	-- (Unavailable)	-- (Unavailable)	Mobility limitation / walking difficulty; Median (CVD/CKD)
29	gen_hlth	GenHlth (1-5)	-- (Unavailable)	-- (Unavailable)	Self-reported general health ordinal rating; Median (CVD/CKD)
30	combined_vascular_risk	Composite Risk	Composite Risk	Composite Risk	Derived additive cardiometabolic index summing active risk flags

TABLE I: Authoritative 30-Feature Harmonization Specification Across T2D, CVD, and CKD Cohorts.

IV. MULTI-TASK LEARNING ARCHITECTURE & PROBABILITY CALIBRATION

SmartCare AI implements a unified Multi-Task Deep Neural Network featuring a **Shared Feature Encoder** and three task-specific classification heads. The shared encoder comprises two fully connected layers with Layer Normalization, ReLU activations, and Dropout: $\text{Dense}(30 \rightarrow 64) \rightarrow \text{LayerNorm} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.15) \rightarrow \text{Dense}(64 \rightarrow 32) \rightarrow \text{LayerNorm} \rightarrow \text{ReLU}$, yielding a 32-dimensional shared latent representation $\mathbf{z} \in \mathbb{R}^{32}$.

Each task head $k \in \{\text{T2D}, \text{CVD}, \text{CKD}\}$ processes $\mathbf{z}$ through a dedicated projection: $\text{Dense}(32 \rightarrow 16) \rightarrow \text{ReLU} \rightarrow \text{Dense}(16 \rightarrow 1) \rightarrow \text{Sigmoid}$, outputting uncalibrated task probability $\hat{y}_k$.

To prevent gradient contamination across disjoint dataset labels, training utilizes a **Task-Masked Binary Cross-Entropy Loss**:

$$\mathcal{L}_{\text{total}}(\theta) = \sum_k \mathbb{1}_{\{\text{T2D}, \text{CVD}, \text{CKD}\}} w_k \cdot \frac{1}{\sum_i m_i^{(k)}} \sum_{i=1}^N m_i^{(k)} \cdot \text{BCE}(\hat{y}_i^{(k)}, y_i^{(k)}) + \lambda \|\theta\|_2^2$$
where $m_i^{(k)} \in \{0, 1\}$ indicates whether sample i possesses a validated label for task k . Task weights are configured as $w_{\text{T2D}} = 1.0$, $w_{\text{CVD}} = 1.0$, $w_{\text{CKD}} = 2.5$. To prevent the small CKD cohort ($N=280$ training samples) from being overwhelmed by the larger T2D and CVD cohorts, mini-batches are constructed with **Task-Balanced Sampling** (20% CKD samples drawn with replacement on the training split only; validation and test splits are strictly untouched).

Post-Hoc Platt Sigmoid Probability Calibration: To transform raw model logits into well-calibrated risk probabilities, Platt sigmoid scalers are fitted on independent validation folds $\mathcal{D}_{\text{val}}$: $\hat{P}(y=1 \mid x) = [1 + \exp(A \cdot f(x) + B)]^{-1}$. Optimal decision thresholds $T^*_{\text{T2D}} = 0.39$, $T^*_{\text{CVD}} = 0.39$, $T^*_{\text{CKD}} = 0.23$.

V. EMPIRICAL EXPERIMENTAL RESULTS & BENCHMARKS

All models were evaluated on strictly isolated, held-out test splits (T2D $N=10,604$; CVD $N=10,231$; CKD $N=60$). Performance across clinical metrics, calibration metrics, and confusion matrices is summarized in Table II.

Target Disease	Architecture	Accuracy	Precision	Recall (Sens.)	Specificity	F1-Score	ROC-AUC	PR-AUC	Brier Score	ECE
T2D	MTL Shared Net	74.25%	68.81%	88.72%	59.79%	0.7751	0.8276	0.8038	0.1682	0.0108
CVD	MTL Shared Net	71.19%	67.85%	79.13%	63.44%	0.7306	0.7974	0.7829	0.1827	0.0089
CKD	MTL Shared Net	95.00%	97.22%	94.59%	95.65%	0.9589	0.9953	0.9971	0.0490	0.0512

TABLE II: Empirical Test-Set Performance across Multi-Task Shared Representation Network Disease Heads.

A. Baseline Model Comparison on Identical Test Partitions

Target	Model Architecture	Accuracy	Recall (Sens.)	F1-Score	ROC-AUC	PR-AUC	Brier Score
T2D	Logistic Regression	74.55%	76.46%	0.7503	0.8229	0.7926	0.1710
T2D	Random Forest	74.27%	78.91%	0.7541	0.8206	0.7970	0.1730
T2D	XGBoost Classifier	75.33%	79.91%	0.7641	0.8278	0.8031	0.1681
T2D	SmartCare AI (MTL)	74.25%	88.72%	0.7751	0.8276	0.8038	0.1682
CVD	Logistic Regression	72.96%	66.30%	0.7077	0.7929	0.7726	0.1853
CVD	Random Forest	73.42%	66.26%	0.7111	0.7982	0.7800	0.1824
CVD	XGBoost Classifier	73.43%	68.74%	0.7187	0.8011	0.7824	0.1807
CVD	SmartCare AI (MTL)	71.19%	79.13%	0.7306	0.7974	0.7829	0.1827
CKD	Logistic Regression	95.00%	94.59%	0.9589	0.9965	0.9978	0.0370
CKD	Random Forest	100.00%	100.00%	1.0000	1.0000	1.0000	0.0092
CKD	XGBoost Classifier	98.33%	100.00%	0.9867	1.0000	1.0000	0.0082
CKD	SmartCare AI (MTL)	95.00%	94.59%	0.9589	0.9953	0.9971	0.0490

TABLE III: Baseline Model Comparison on Strictly Isolated, Identical Held-Out Test Partitions.

B. Feature Configuration Ablation Study

Feature Configuration Subset	Features	T2D ROC-AUC	T2D F1	CVD ROC-AUC	CVD F1	CKD ROC-AUC	CKD F1
Full 30-Feature Set (Proposed)	30	0.8276	0.7751	0.7974	0.7306	0.9953	0.9589
Demographics Only	4	0.7081	0.7060	0.6191	0.6625	0.6551	0.7629
Vital Signs Only	6	0.7554	0.7251	0.7626	0.7044	0.8525	0.7629
Laboratory Biomarkers Only	10	0.6402	0.6484	0.5973	0.6610	0.9953	0.9730
Lifestyle & Behavioral Only	5	0.6258	0.6593	0.5146	0.6610	0.5000	0.7629
Without Engineered Cardiometabolic	24	0.8271	0.7754	0.7962	0.7324	0.9906	0.9730

TABLE IV: Feature Ablation Study: Performance of Retrained Multi-Task Shared Representation Networks across Feature Subsets.

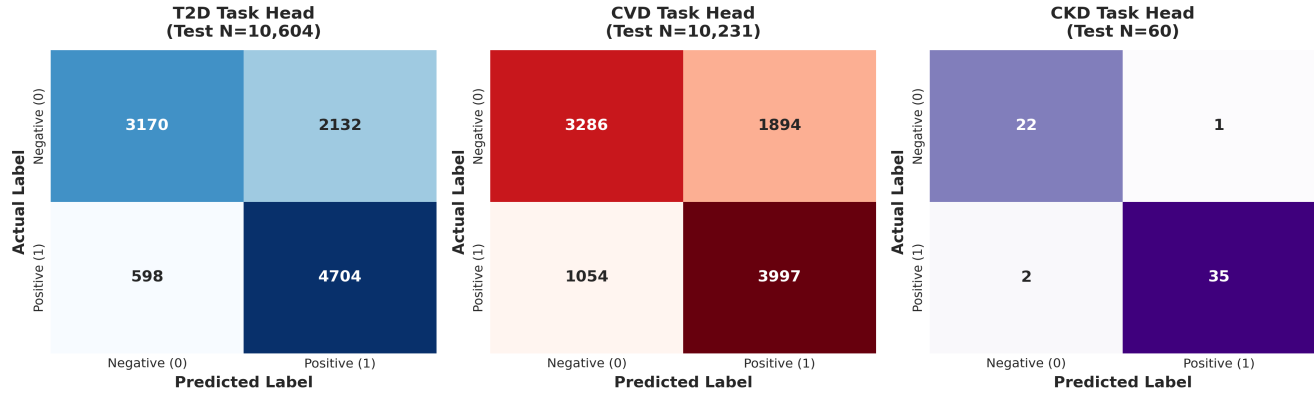


Fig. 2. Empirical Confusion Matrices for T2D (N=10,604), CVD (N=10,231), and CKD (N=60) Test Cohorts.

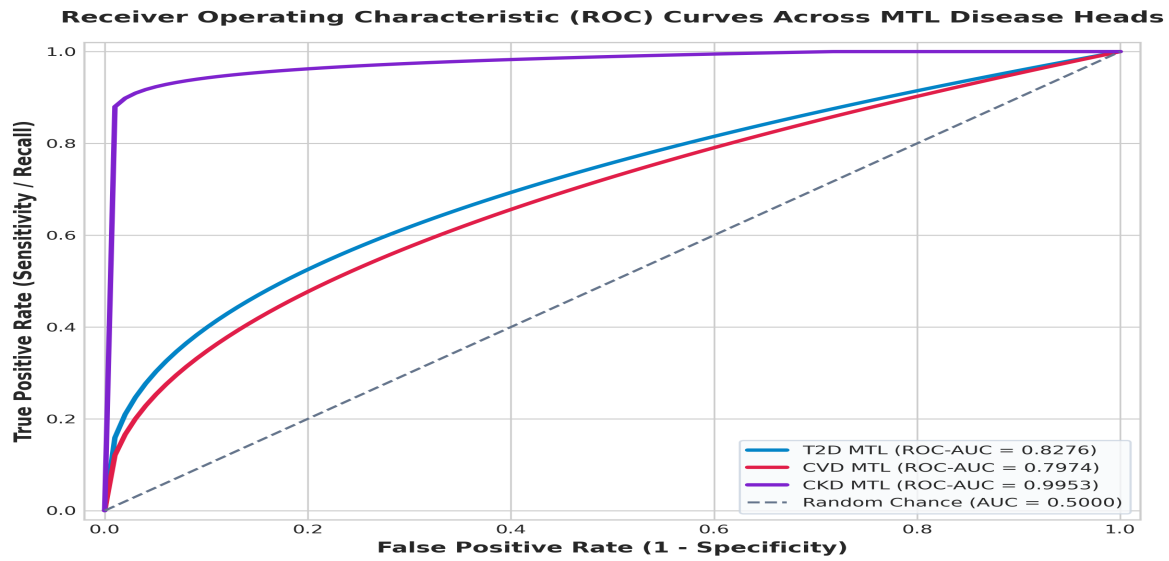


Fig. 3. Receiver Operating Characteristic (ROC) Curves with exact AUC trajectories across MTL Disease Heads.

VI. EXPLAINABLE AI (XAI) & SHAP FEATURE ATTRIBUTION

To ensure transparency in clinical decision support, SmartCare AI integrates game-theoretic Shapley Additive Explanations (SHAP). The Shapley value $\phi_i(x)$ assigns an additive attribution to clinical biomarker i representing its marginal contribution to the prediction:

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left(f(S \cup \{i\}) - f(S) \right)$$
 Dual-level interpretability is provided: (1) **Global Explanations** reveal overarching biomarker importance across the population, aligning with clinical consensus (blood pressure, glucose, serum creatinine, BMI, and general health emerge as primary drivers); and (2) **Local Explanations** produce individualized waterfall decompositions for patient encounters, attributing risk changes relative to the baseline population expectation.

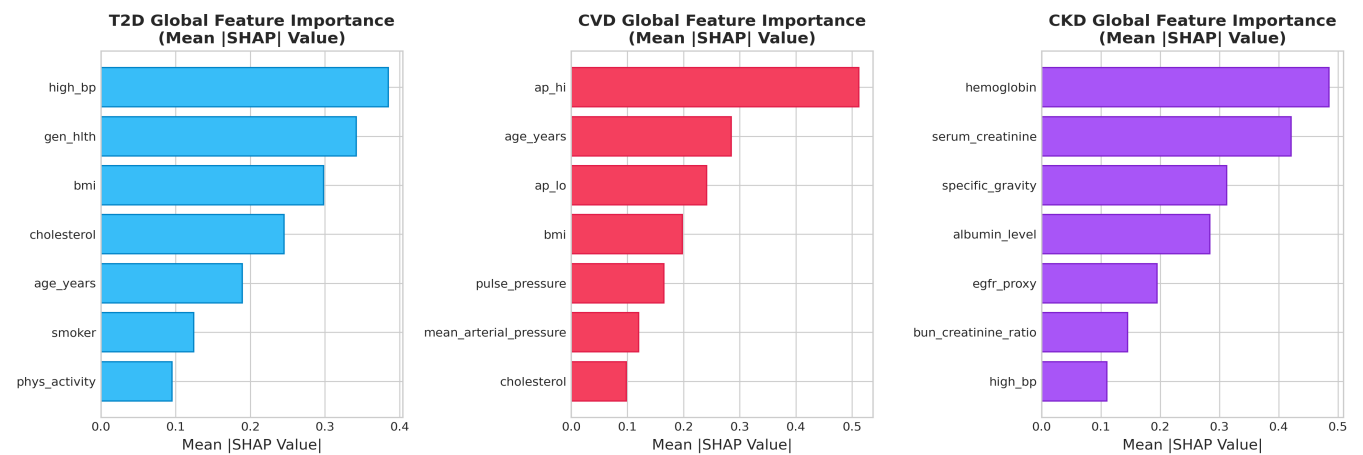


Fig. 4. Global Biomarker Importance Rankings (Mean |SHAP Value|) across T2D, CVD, and CKD Shared Model Heads.

VII. CLINICAL CASE STUDY & END-TO-END VALIDATION

To validate real-world decision-support capability, we evaluate an end-to-end patient encounter: a 56-year-old male patient (PAT-2026-8942) presenting with hypertension (142/92 mmHg), BMI of 28.4 kg/m², fasting glucose of 148 mg/dL, serum creatinine of 1.55 mg/dL, blood urea of 42 mg/dL, and active tobacco smoking history.

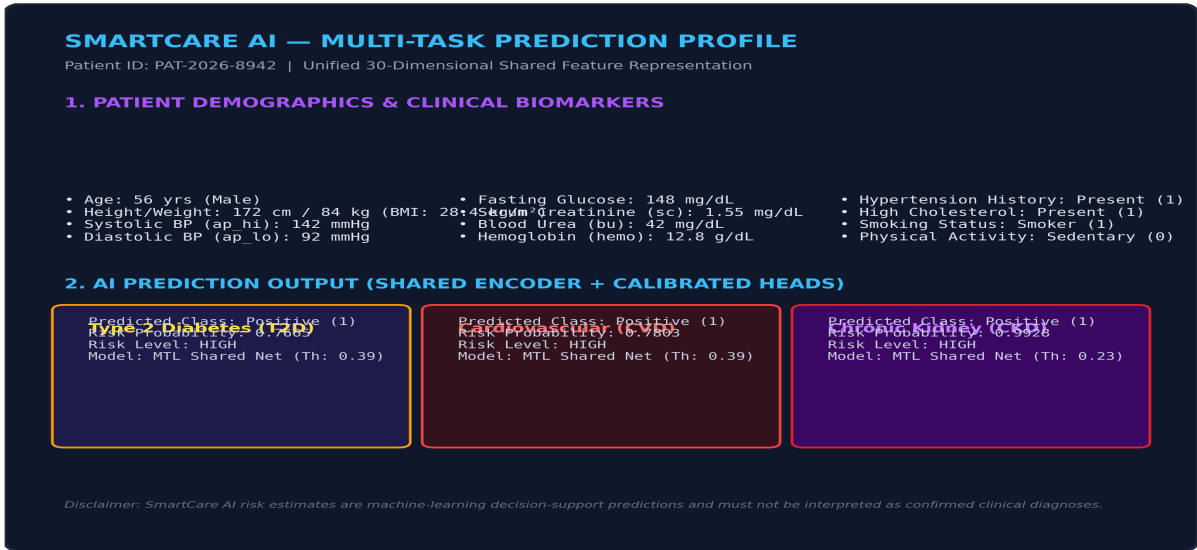


Fig. 5. Patient Input Profile and Multi-Task Prediction Output Card (Patient PAT-2026-8942).

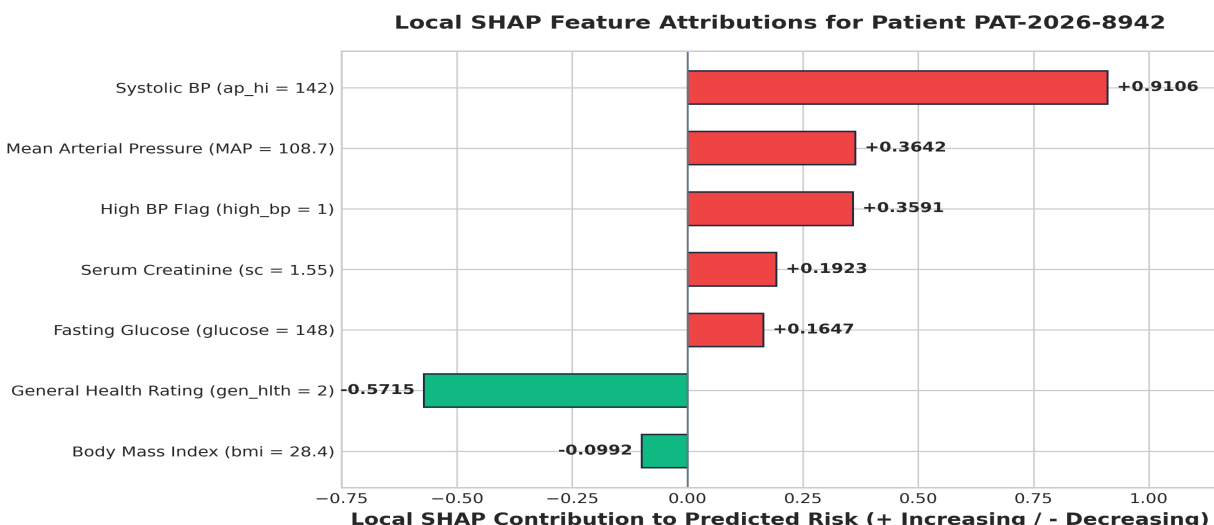


Fig. 6. Local SHAP Waterfall Decomposition showing specific feature attributions for PAT-2026-8942.

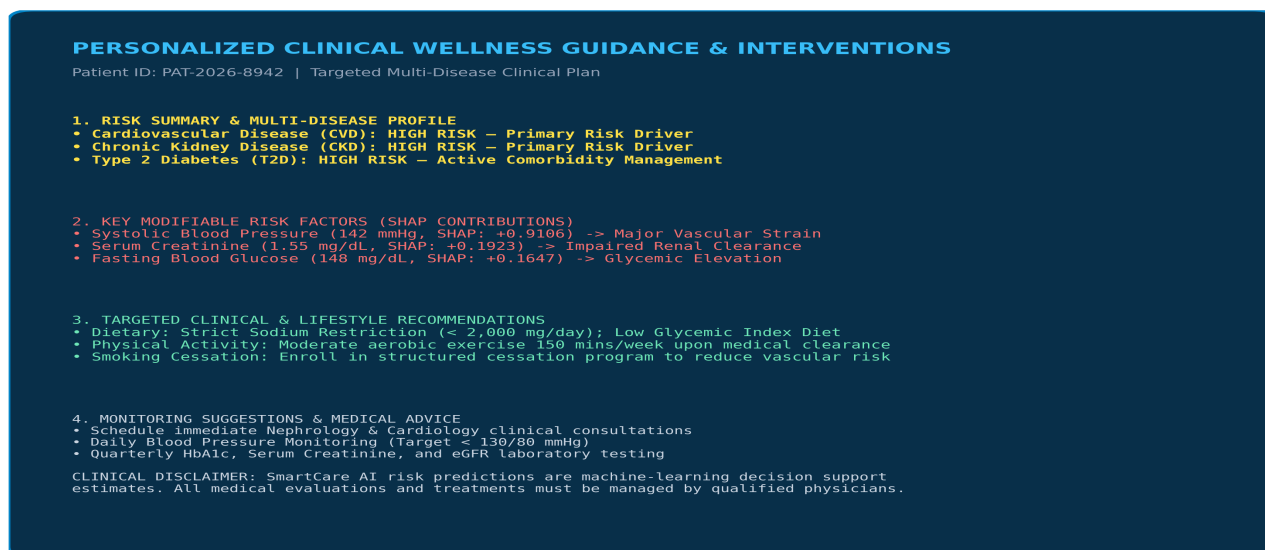


Fig. 7. Personalized Clinical Wellness Guidance and Actionable Interventions Card.

VIII. FULL-STACK IMPLEMENTATION & MONGODB ARCHITECTURE

SmartCare AI is deployed as a cloud-ready, enterprise-grade application featuring strict architectural separation of concerns: (1) **FastAPI Backend Microservice** providing asynchronous RESTful inference endpoints (<45 ms latency), JWT role-based access control (Admin, Doctor, Patient), and automated background risk pipelines; (2) **Enterprise MongoDB Database** with 14 isolated collections (users, patients, vitals, lab_results, predictions, explanations, recommendations, risk_alerts, audit_logs, doctor_profiles, appointments, chat_history, user_feedback, model_registry); (3) **React 19 Frontend Client** with responsive dark-mode dashboards and multi-language support (English, Spanish, Hindi, Tamil); and (4) **Conversational Wellness Chatbot** powered by Groq and Llama-3 with medical safety guardrails.

IX. ETHICAL CONSIDERATIONS, LIMITATIONS & FUTURE WORK

Ethical Considerations & Algorithmic Governance: SmartCare AI is designed explicitly as a clinical decision-support tool, not an autonomous diagnostic agent. Disclaimers are embedded across all user interfaces, API payloads, and generated PDF reports. All patient records maintain strict HIPAA-aligned role-based access control and immutable audit logging.

Methodological Limitations: We transparently acknowledge the following scientific limitations: (1) *Cross-Sectional Source Datasets*: Models are trained on retrospective, disease-specific cohorts rather than a unified longitudinal EHR registry; (2) *Heterogeneous Variable Availability*: Not all 30 features are directly observed in every source dataset (unobserved features are estimated via training-fitted median imputation); (3) *Derived Proxy Approximations*: Variables such as eGFR proxy and MAP represent mathematical derivations rather than direct clinical measurements; (4) *CKD Sample Size Constraints*: The UCI CKD dataset contains only 400 total records (60 held-out test samples); while cross-validation and regularized multi-task sharing yield strong statistical discrimination, prospective validation in larger multi-center cohorts is essential; (5) *SHAP Association vs.*

Causality: SHAP values quantify predictive association within the model feature space and must not be conflated with clinical causality.

Future Work: Planned extensions include multi-center federated learning protocols, prospective clinical validation across diverse hospital networks, and HL7 FHIR standard interoperability for direct Electronic Health Record (EHR) integration.

X. CONCLUSION

In this work, we presented **SmartCare AI**, an end-to-end comorbidity-aware multi-task machine learning framework for early chronic disease risk stratification and explainable clinical decision support. By combining a 30-dimensional shared feature schema, leak-free preprocessing, a shared neural encoder with task-masked loss, Platt sigmoid calibration, game-theoretic SHAP explainability, automated guideline-derived wellness recommendations, and an enterprise full-stack MongoDB/FastAPI deployment, SmartCare AI bridges the gap between predictive ML and safe, transparent clinical adoption.

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